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1 AI-Based Intrusion Detection Systems for Next-Generation Networks

1.1 Complete PhD Research Proposal

Candidate: idriss5214

Date: December 22, 2025

Duration: 4 Years

Field: Computer Science - Cybersecurity & Network Security

1.2 1. Executive Summary

The rapid evolution of next-generation networks (NGN), including 5G/6G, Internet of Things (IoT), Software-Defined Networks (SDN), and edge computing, has introduced unprecedented security challenges. Traditional intrusion detection systems (IDS) are inadequate for detecting sophisticated cyber threats in these complex, dynamic environments. This PhD research proposes to develop an advanced, AI-based intrusion detection framework that leverages hybrid deep learning models, adaptive learning mechanisms, and explainable AI to provide real-time, accurate, and transparent threat detection across diverse network architectures.

The proposed research will integrate Convolutional Neural Networks (CNN), Long Short-Term Memory (LSTM) networks, Transformers, and ensemble methods to create a robust detection engine capable of identifying both known and zero-day attacks. The system will incorporate reinforcement learning for adaptive threat response, federated learning for privacy-preserving distributed detection, and explainable AI (XAI) techniques to ensure transparency and trust in automated security decisions.

Key Innovations: - Hybrid CNN-LSTM-Transformer architecture for multi-scale feature extraction - Adaptive learning framework using reinforcement learning (RL) and online learning - Distributed IDS architecture for edge computing with federated learning - Explainable AI integration for transparent decision-making - Real-time detection with <100ms latency for critical network scenarios - Privacy-preserving detection through federated and differential privacy mechanisms

Expected Impact: - Improved detection accuracy (>98%) with reduced false positive rates (<1%) - Scalable solution for heterogeneous NGN environments - Enhanced security for critical infrastructure and IoT ecosystems - Practical deployment frameworks for industry adoption - 5-10 high-impact publications in top security venues

1.3 2. Introduction

1.3.1 2.1 Background

Next-generation networks represent a paradigm shift in telecommunications and computing infrastructure. 5G/6G networks promise ultra-low latency (<1ms), massive connectivity (1M+ devices/km²), and multi-gigabit speeds. Software-Defined Networks (SDN) and Network Function Virtualization (NFV) enable flexible, programmable network management. Edge computing brings computation closer to data sources, reducing latency and bandwidth consumption. The Internet of Things is connecting billions of resource-constrained devices across diverse applications from smart homes to industrial control systems.

However, these advancements introduce significant security vulnerabilities:

1. **Expanded Attack Surface:** Billions of IoT devices, many with weak security, create numerous entry points
2. **Network Complexity:** SDN/NFV introduce new attack vectors in control plane and virtualization layers
3. **Heterogeneity:** Diverse device types, protocols, and traffic patterns complicate detection
4. **High-Speed Traffic:** Gigabit speeds require real-time detection with minimal latency
5. **Evolving Threats:** Advanced persistent threats (APTs) and zero-day exploits evade signature-based detection
6. **Resource Constraints:** Edge and IoT devices have limited computational resources for security
7. **Privacy Concerns:** Centralized IDS raise data privacy and sovereignty issues

1.3.2 2.2 Problem Statement

Current intrusion detection systems face critical limitations in next-generation network environments:

Traditional IDS Limitations: - Signature-based systems cannot detect zero-day attacks - Anomaly-based systems generate high false positive rates - Rule-based systems lack adaptability to evolving threats - Centralized architectures create single points of failure - Black-box ML models lack explainability for security analysts - Static models fail to adapt to concept drift in network behavior - Privacy violations through centralized data collection

Research Challenges: 1. How to achieve high detection accuracy (>98%) while maintaining low false positive rates (<1%)? 2. How to detect zero-day attacks and advanced persistent threats without prior signatures? 3. How to process high-speed network traffic in real-time with <100ms detection latency? 4. How to create scalable IDS for distributed edge computing environments? 5. How to ensure model explainability and trust in automated security decisions? 6. How to preserve data privacy while enabling collaborative threat intelligence? 7. How to adapt detection models continuously to evolving threat landscapes?

1.3.3 2.3 Research Significance

This research addresses critical gaps at the intersection of artificial intelligence, cybersecurity, and next-generation networks. The proposed AI-based IDS will:

- **Academic Contribution:** Advance the state-of-art in AI-driven network security through novel hybrid architectures, adaptive learning frameworks, and distributed detection mechanisms
 - **Practical Impact:** Provide deployable solutions for securing critical infrastructure, enterprise networks, IoT ecosystems, and telecommunications providers
 - **Societal Benefit:** Enhance security and privacy for billions of users in an increasingly connected world
 - **Economic Value:** Reduce cybercrime costs (projected \$10.5T annually by 2025) through improved threat prevention
-

1.4 3. Research Objectives

1.4.1 3.1 Primary Objective

To design, develop, and validate an advanced AI-based intrusion detection system for next-generation networks that achieves superior detection accuracy, real-time performance, scalability, adaptability, and explainability compared to existing solutions.

1.4.2 3.2 Specific Objectives

1. Hybrid AI Model Development

- Design CNN-LSTM hybrid for spatial-temporal feature extraction

- Implement Transformer networks for capturing long-range dependencies
 - Develop autoencoder architectures for unsupervised anomaly detection
 - Create ensemble methods combining multiple AI models
2. **Adaptive Learning Framework**
 - Implement reinforcement learning (DQN, A3C, PPO) for dynamic threat response
 - Design online learning mechanisms for continuous model updates
 - Develop transfer learning strategies for cross-domain attack detection
 - Create meta-learning approaches for rapid adaptation to new threats
 3. **Distributed IDS Architecture**
 - Design hierarchical edge-fog-cloud detection framework
 - Implement federated learning for privacy-preserving collaborative detection
 - Develop efficient model compression for resource-constrained devices
 - Create secure aggregation protocols for distributed learning
 4. **Explainable AI Integration**
 - Integrate SHAP, LIME, and attention mechanisms for model interpretability
 - Develop visualization tools for security analysts
 - Create explainability metrics for IDS evaluation
 - Design human-in-the-loop feedback mechanisms
 5. **Real-Time Optimization**
 - Optimize detection latency to <100ms for critical scenarios
 - Implement efficient feature extraction and model inference
 - Design hardware acceleration strategies (GPU, FPGA)
 - Develop traffic prioritization and sampling techniques
 6. **Comprehensive Validation**
 - Evaluate on multiple benchmark datasets (NSL-KDD, CICIDS2017, UNSW-NB15, IoT-23)
 - Deploy testbed with real 5G/IoT/SDN infrastructure
 - Conduct extensive experiments across diverse attack scenarios
 - Compare against state-of-the-art IDS solutions
 7. **Privacy and Security**
 - Implement differential privacy mechanisms
 - Design secure multi-party computation protocols
 - Address adversarial machine learning threats
 - Ensure GDPR and data protection compliance
-

1.5 4. Literature Review

1.5.1 4.1 Traditional Intrusion Detection Systems

Signature-Based IDS: Traditional systems like Snort and Suricata rely on pattern matching against known attack signatures. While effective for known threats with low false positives, they fail to detect zero-day attacks and polymorphic malware [1]. The signature database requires constant updates, creating maintenance overhead and detection gaps.

Anomaly-Based IDS: Statistical and machine learning approaches establish baseline normal behavior and flag deviations as anomalies [2]. Classical ML techniques (SVM, Random Forest, k-NN) achieve moderate success but suffer from high false positive rates (10-20%) and inability to adapt to concept drift [3].

Hybrid Approaches: Systems combining signature and anomaly detection (e.g., OS-SEC, Bro/Zeek) provide better coverage but inherit limitations from both approaches [4]. Computational overhead increases significantly with hybrid implementations.

1.5.2 4.2 AI/ML Techniques in Cybersecurity

Classical Machine Learning: Random Forest, SVM, and ensemble methods have been extensively studied for intrusion detection [5, 6]. While achieving 90-95% accuracy on benchmark datasets, these methods struggle with high-dimensional feature spaces and require extensive feature engineering.

Deep Learning: CNN architectures extract spatial features from network traffic representations [7]. LSTM networks model temporal sequences for behavior analysis [8]. Autoencoder-based anomaly detection identifies outliers in unsupervised settings [9]. Deep learning achieves 95-98% accuracy but lacks interpretability.

Ensemble Methods: Combining multiple models improves robustness and accuracy [10]. Stacking, bagging, and boosting techniques have shown 2-5% accuracy improvements over individual models.

Reinforcement Learning: RL agents learn optimal defense strategies through interaction with network environments [11]. Q-learning and Deep Q-Networks enable adaptive response to evolving threats but require extensive training.

1.5.3 4.3 Next-Generation Network Security

5G/6G Security: Network slicing, edge computing, and massive IoT connectivity introduce new vulnerabilities [12]. ML-based anomaly detection for 5G core networks shows promise but faces scalability challenges [13].

IoT Security: Resource-constrained IoT devices require lightweight detection mechanisms [14]. Federated learning enables collaborative threat detection without centralizing sensitive data [15].

SDN/NFV Security: Software-defined architectures face controller compromise and flow rule manipulation attacks [16]. ML-based SDN security solutions achieve real-time threat detection at controller level [17].

Edge Computing: Distributed detection at edge nodes reduces latency and bandwidth but requires model compression and efficient aggregation [18].

1.5.4 4.4 Recent Advances (2019-2025)

Transformer Networks: Attention mechanisms capture long-range dependencies in network traffic sequences [19]. BERT-like architectures for IDS achieve state-of-the-art results on benchmark datasets [20].

Federated Learning: Privacy-preserving collaborative learning enables distributed IDS without data centralization [21]. Secure aggregation protocols protect against adversarial participants [22].

Explainable AI: SHAP and LIME integration provides feature-level explanations for ML predictions [23]. Attention visualization in deep models reveals decision rationale [24].

Graph Neural Networks: GNNs model network topology and traffic flows as graphs, capturing structural patterns [25]. Application to SDN security shows promising results [26].

Adversarial Machine Learning: Adversarial training improves robustness against evasion attacks [27]. Defensive distillation and certified defenses enhance IDS resilience [28].

1.5.5 4.5 Research Gaps

1. **Limited Hybrid Architectures:** Few studies combine CNN, LSTM, and Transformers effectively
 2. **Inadequate Real-Time Performance:** Existing deep learning IDS struggle with <100ms latency requirements
 3. **Scalability Issues:** Most solutions tested on small-scale networks, not production-level deployments
 4. **Explainability Gap:** Black-box models lack transparency needed for security operations
 5. **Privacy Concerns:** Centralized architectures violate data sovereignty requirements
 6. **Adaptation Limitations:** Static models fail to handle concept drift and evolving threats
 7. **Cross-Domain Transfer:** Limited research on transferring IDS models across different network types
 8. **Benchmark Limitations:** Existing datasets don't adequately represent modern 5G/IoT attack scenarios
-

1.6 5. Research Methodology

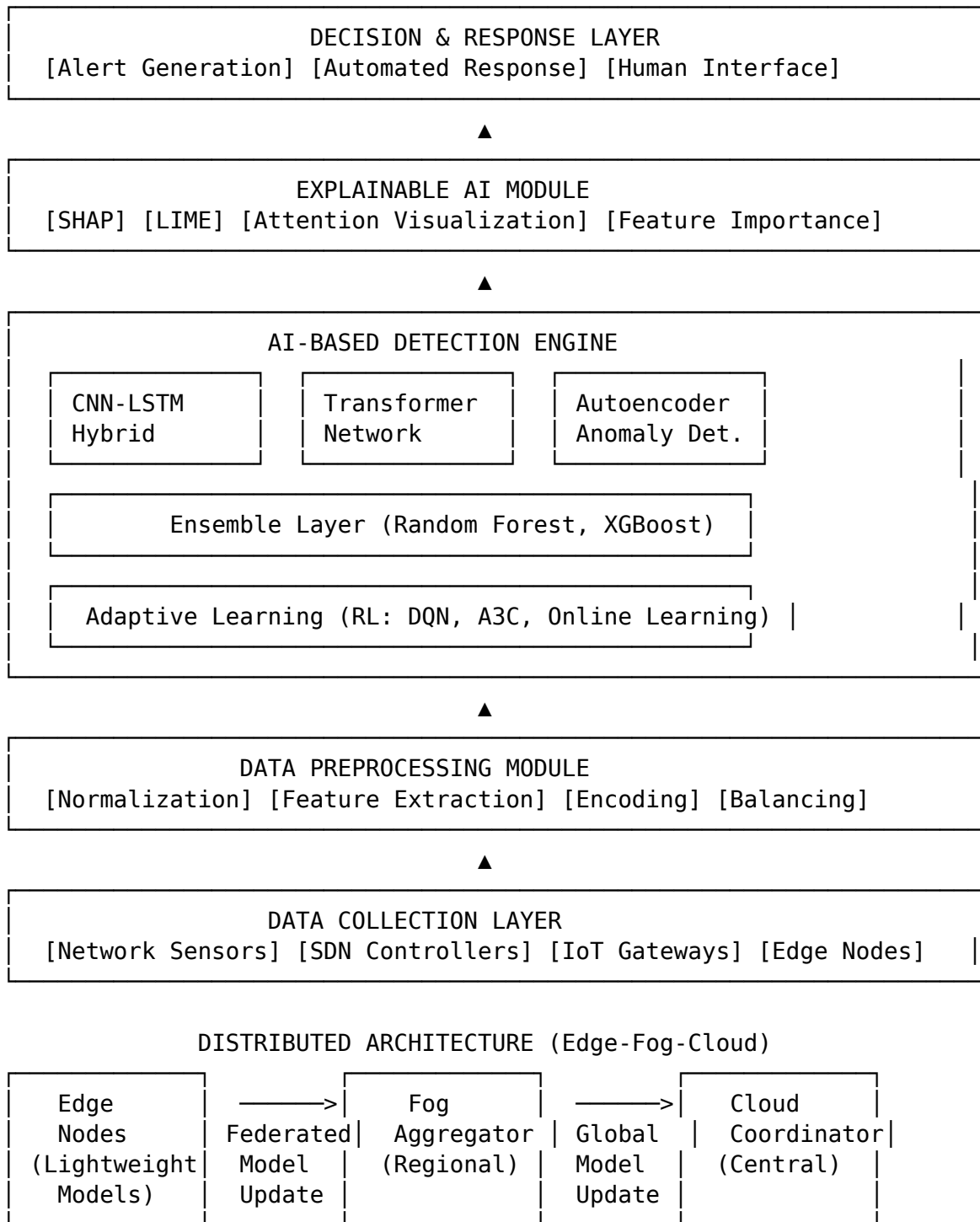
1.6.1 5.1 Research Design

This research follows a design science approach with iterative development, evaluation, and refinement cycles. The methodology combines theoretical modeling, algorithm design, system implementation, and empirical validation.

Phases: 1. **Phase 1 (Months 1-6):** Literature review, requirements analysis, initial design 2. **Phase 2 (Months 7-12):** Algorithm development, prototype implementation 3. **Phase 3 (Months 13-24):** Framework integration, testbed deployment, initial experiments 4. **Phase 4 (Months 25-36):** Extensive validation, optimization, publications 5. **Phase 5 (Months 37-48):** Final experiments, thesis writing, defense preparation

1.6.2 5.2 Framework Architecture

The proposed AI-IDS framework consists of seven interconnected layers:



1.6.3 5.3 Key Components

1.6.3.1 5.3.1 Data Collection Layer

- **Network Sensors:** Deep packet inspection, flow monitoring, protocol analysis
- **SDN Controllers:** OpenFlow statistics, topology information, control plane events
- **IoT Gateways:** Device behavior monitoring, protocol translation, edge preprocessing
- **Log Aggregation:** System logs, application logs, security events

1.6.3.2 5.3.2 Data Preprocessing Module

- **Traffic Parsing:** Extract relevant features from raw packets/flows
- **Feature Engineering:** Statistical features, temporal features, behavioral features
- **Normalization:** Min-max scaling, Z-score normalization
- **Encoding:** One-hot encoding for categorical features, embedding for high-cardinality
- **Balancing:** SMOTE, ADASYN for addressing class imbalance
- **Dimensionality Reduction:** PCA, autoencoders for feature compression

1.6.3.3 5.3.3 AI-Based Detection Engine CNN-LSTM Hybrid: - CNN layers extract spatial features from traffic matrices - LSTM layers model temporal dependencies in traffic sequences - Attention mechanism focuses on relevant time steps - Architecture: Conv1D(64) → Conv1D(128) → LSTM(128) → LSTM(64) → Dense(num_classes)

Transformer Network: - Multi-head self-attention captures global dependencies - Position encoding preserves temporal information - Feed-forward networks for feature transformation - Architecture: Embedding → Positional Encoding → 6× Transformer Blocks → Classification Head

Autoencoder for Anomaly Detection: - Encoder compresses normal traffic into latent representation - Decoder reconstructs input from latent space - Reconstruction error threshold for anomaly detection - Variational autoencoder for probabilistic modeling

Ensemble Layer: - Random Forest (100 trees) for robustness - XGBoost for gradient boosting - Voting or stacking for final prediction - Confidence calibration for uncertainty quantification

Adaptive Learning: - Deep Q-Network (DQN) for optimal response selection - Actor-Critic (A3C) for policy-based learning - Proximal Policy Optimization (PPO) for stable training - Online learning for continuous model updates with streaming data

1.6.3.4 5.3.4 Explainable AI Module

- **SHAP:** Shapley values for global and local feature importance
- **LIME:** Local surrogate models for instance-level explanations

- **Attention Visualization:** Heatmaps showing attended traffic segments
- **Counterfactual Explanations:** What-if analysis for decision understanding
- **Rule Extraction:** Distilling neural networks into interpretable rules

1.6.3.5 5.3.5 Distributed Architecture

- **Edge Nodes:** Lightweight models (MobileNet-like) for local detection
- **Fog Layer:** Regional aggregation and model coordination
- **Cloud Layer:** Global model training and distribution
- **Federated Learning:** Privacy-preserving collaborative learning
- **Model Compression:** Quantization, pruning, knowledge distillation

1.6.4 5.4 Datasets

Dataset	Description	Samples	Features	Attack Types
NSL-KDD	Refined version of KDD'99, balanced classes	125,973 train, 22,544 test	41	DoS, Probe, R2L, U2R
CICIDS2017	Realistic modern attacks, labeled flows	2.8M flows	80	DoS, DDoS, PortScan, Brute Force, Web attacks, Botnet
UNSW-NB15	Hybrid of real and synthetic attacks	2.54M records	49	Fuzzers, Analysis, Backdoors, DoS, Exploits, Generic, Reconnaissance, Shellcode, Worms
IoT-23	IoT-specific botnet traffic	325M packets	Variable	Mirai, Torii, IoT-specific malware
5G Dataset	Custom collection from testbed	TBD	TBD	5G-specific attacks, network slicing attacks

Data Preprocessing Pipeline: 1. Traffic capture using tcpdump/Wireshark 2. Flow generation using CICFlowMeter 3. Feature extraction (statistical, temporal, protocol-based) 4. Cleaning (missing values, outliers) 5. Normalization and encoding 6. Train/validation/test split (70/15/15) 7. Class balancing using SMOTE for minority classes

1.6.5 5.5 Experimental Setup

Hardware: - GPU Cluster: 4× NVIDIA A100 (40GB) for deep learning training - CPU Servers: 2× Intel Xeon Gold 6248R (48 cores) for preprocessing - Edge Devices: 10× Raspberry Pi 4 (8GB) for edge testing - SDN Testbed: 5× OpenFlow switches, ONOS controller - IoT Devices: 50× mixed IoT sensors and actuators

Software: - Deep Learning: TensorFlow 2.x, PyTorch 2.x, Keras - SDN: ONOS, OpenDaylight, Ryu - Network Simulation: NS-3, Mininet, OMNET++ - XAI Tools: SHAP, LIME, Captum - Federated Learning: TensorFlow Federated, PySyft - Data Processing: Pandas, NumPy, Scikit-learn - Monitoring: Grafana, Prometheus, ELK Stack

Baselines for Comparison: 1. Traditional ML: Random Forest, SVM, Decision Trees 2. Deep Learning: CNN, LSTM, GRU, standalone models 3. Existing IDS: Snort, Suricata, Zeek 4. Recent Research: Latest published models from top conferences

1.6.6 5.6 Evaluation Metrics

Detection Performance:

Metric	Formula	Target
Accuracy	$(TP + TN) / (TP + TN + FP + FN)$	>98%
Precision	$TP / (TP + FP)$	>97%
Recall/TPR	$TP / (TP + FN)$	>98%
F1-Score	$2 \times (Precision \times Recall) / (Precision + Recall)$	>97%
False Positive Rate	$FP / (FP + TN)$	<1%
AUC-ROC	Area under ROC curve	>0.99

Efficiency Metrics: - Detection Latency: Average time from packet arrival to detection decision (<100ms) - Throughput: Number of packets processed per second (>10K pps) - CPU Usage: Average CPU utilization during detection (<50%) - Memory Footprint: RAM consumption (<2GB for edge devices) - Network Overhead: Additional bandwidth for IDS communication (<5%)

Scalability Metrics: - Performance vs. Network Size (100 to 100K devices) - Distributed efficiency (speedup, communication cost) - Model update latency in federated setting

Adaptability Metrics: - Zero-day detection rate: % of unknown attacks detected - Adaptation time: Time to adapt to new attack patterns - Concept drift handling: Performance degradation over time

Explainability Metrics: - Feature importance consistency - Explanation fidelity (faithfulness to model) - Human evaluation scores (clarity, usefulness)

1.7 6. Expected Contributions

1.7.1 6.1 Theoretical Contributions

1. **Novel Hybrid Architecture:** CNN-LSTM-Transformer fusion with attention mechanisms for multi-scale feature extraction in network traffic analysis
2. **Adaptive Learning Framework:** Reinforcement learning-based dynamic threat response with online learning for continuous adaptation

3. **Distributed Detection Theory:** Mathematical frameworks for federated IDS optimization with privacy guarantees
4. **Explainability Models:** New XAI techniques specifically designed for network intrusion detection
5. **Transfer Learning Strategies:** Cross-domain knowledge transfer methods for heterogeneous network environments

1.7.2 6.2 Practical Contributions

1. **Production-Ready Framework:** Open-source AI-IDS platform deployable in real networks
 - GitHub repository with comprehensive documentation
 - Docker containers for easy deployment
 - APIs for integration with existing security infrastructure
2. **Benchmark Datasets:** Curated and preprocessed datasets for reproducible research
 - Modern 5G/IoT attack scenarios
 - Balanced and labeled for training/evaluation
3. **Industry Guidelines:** Best practices for deploying AI-based IDS in production
 - Performance tuning recommendations
 - Security and privacy considerations
 - Maintenance and update procedures
4. **Tool Ecosystem:** Suite of tools for IDS development and evaluation
 - Automated feature engineering pipeline
 - Model training and evaluation framework
 - Explainability dashboard for analysts
5. **Testbed Infrastructure:** Documented setup for 5G/IoT/SDN security testbed
 - Replicable for other research institutions
 - Virtual testbed configurations for cloud deployment

1.8 7. Challenges and Risk Mitigation

Challenge	Risk Level	Mitigation Strategy
Data Availability	High	Use multiple public datasets; establish industry partnerships for real traffic; generate synthetic data using GANs
Computational Resources	Medium	Apply for AWS/Google Cloud research credits; use model compression; leverage university GPU clusters
Model Overfitting	Medium	Cross-validation, regularization (L2, dropout), early stopping; test on multiple datasets

Challenge	Risk Level	Mitigation Strategy
Real-Time Performance	High	Model optimization (pruning, quantization); hardware acceleration; efficient preprocessing
Adversarial Attacks	Medium	Adversarial training; defensive distillation; input validation; ensemble diversity
Concept Drift	High	Online learning implementation; periodic retraining; drift detection mechanisms
Explainability Trade-off	Medium	Balance between accuracy and interpretability; multiple XAI techniques; user studies
Privacy Concerns	High	Differential privacy; federated learning; secure aggregation; compliance with regulations
Testbed Limitations	Medium	Hybrid approach (simulation + physical); cloud-based virtual testbed; phased deployment
Industry Adoption	Low	Focus on practical usability; documentation; workshops; open-source release

1.9 8. Timeline

1.9.1 Year 1: Foundation and Design (Months 1-12)

Q1 (Months 1-3): Literature Review & Coursework - Comprehensive literature review (100+ papers) - PhD coursework (Advanced ML, Network Security) - Form dissertation committee - Preliminary experiments with baseline models - **Deliverable:** Literature review draft, committee approval

Q2 (Months 4-6): Algorithm Design & Initial Prototyping - Design CNN-LSTM hybrid architecture - Implement baseline models (RF, SVM, standalone CNN/LSTM) - Collect and preprocess benchmark datasets - Set up development environment - **Deliverable:** Algorithm specifications, initial results

Q3 (Months 7-9): Transformer & Ensemble Implementation - Implement Transformer-based IDS - Develop autoencoder for anomaly detection - Create ensemble methods - Preliminary evaluation on NSL-KDD and CICIDS2017 - **Deliverable:** Prototype models, technical report

Q4 (Months 10-12): Comprehensive Exam & First Publication - Prepare for comprehensive exam - Write first conference paper - Integrate XAI components (SHAP, LIME) - Initial comparison with baselines - **Deliverable:** Comprehensive exam passed, paper submitted, annual report

1.9.2 Year 2: Framework Development (Months 13-24)

Q1 (Months 13-15): Adaptive Learning Integration - Implement reinforcement learning (DQN, A3C) - Develop online learning mechanisms - Transfer learning experiments - Extended evaluation on UNSW-NB15 - **Deliverable:** Adaptive learning module, workshop paper

Q2 (Months 16-18): Distributed Architecture Design - Design federated learning framework - Implement model compression techniques - Develop edge deployment strategy - Privacy mechanism integration - **Deliverable:** Distributed IDS prototype

Q3 (Months 19-21): Testbed Deployment Preparation - Acquire hardware (edge devices, SDN switches) - Set up SDN testbed with ONOS - Deploy IoT devices - Configure monitoring infrastructure - **Deliverable:** Operational testbed

Q4 (Months 22-24): Initial Testbed Experiments - Deploy IDS on testbed - Conduct real-time detection experiments - Performance profiling and optimization - Second major publication - **Deliverable:** Testbed results, journal paper submitted, annual report

1.9.3 Year 3: Validation and Optimization (Months 25-36)

Q1 (Months 25-27): Extensive Experiments - Large-scale experiments on all datasets - Cross-dataset evaluation - Ablation studies on model components - Adversarial robustness testing - **Deliverable:** Comprehensive experimental results

Q2 (Months 28-30): 5G/IoT Specific Validation - Custom 5G dataset collection - IoT-23 malware detection experiments - Edge computing performance analysis - Real-world case studies - **Deliverable:** NGN-specific results, conference paper

Q3 (Months 31-33): Performance Optimization - Model compression and acceleration - Latency optimization (<100ms target) - Scalability experiments (1K to 100K devices) - Energy efficiency analysis - **Deliverable:** Optimized framework

Q4 (Months 34-36): Publication Push & Framework Release - Submit major journal paper (IEEE TIFS/TMC) - Write conference papers for top venues - Prepare open-source release - Documentation and tutorials - **Deliverable:** Framework v1.0 released, multiple papers submitted, annual report

1.9.4 Year 4: Thesis Completion (Months 37-48)

Q1 (Months 37-39): Additional Experiments - Address reviewer feedback from publications - Conduct additional experiments as needed - User studies for explainability evaluation - Industry validation with partners - **Deliverable:** Revised manuscripts, additional results

Q2 (Months 40-42): Thesis Writing - Write dissertation chapters - Integrate all results and publications - Create comprehensive documentation - Prepare figures and tables - **Deliverable:** Thesis draft

Q3 (Months 43-45): Thesis Review & Defense Preparation - Committee review of dissertation - Revisions based on feedback - Prepare defense presentation - Practice

defense - **Deliverable:** Final thesis draft, defense slides

Q4 (Months 46-48): Defense & Final Submission - Dissertation defense - Final revisions - Submit final dissertation - Publish remaining papers - **Deliverable:** PhD degree, published papers, graduated!

Key Milestones: - Month 3: Committee formed - Month 12: Comprehensive exam passed - Month 18: First major publication accepted - Month 24: Testbed operational, second publication submitted - Month 30: 3+ publications accepted/published - Month 36: Framework released, journal paper submitted - Month 42: Thesis draft complete - Month 48: Defense passed, PhD awarded

1.10 9. Resources Required

1.10.1 9.1 Computational Resources

Cloud Computing: - AWS/Google Cloud/Azure research credits: \$10,000-15,000 - GPU instances (p3.8xlarge equivalent): 10,000 hours - Storage: 5TB for datasets and models

University Resources: - Access to HPC cluster with GPU nodes - Dedicated development server - Backup and archival storage

1.10.2 9.2 Software and Tools

Commercial Licenses: - MATLAB (if needed): \$500/year - Professional network analysis tools: \$1,000-2,000

Open Source (Free): - TensorFlow, PyTorch, Keras - Scikit-learn, Pandas, NumPy - ONOS, OpenDaylight, Mininet - SHAP, LIME, Captum

1.10.3 9.3 Network Testbed

Hardware: - 5× OpenFlow-enabled switches: \$5,000 - 10× Raspberry Pi 4 (8GB): \$1,000 - 50× IoT devices (various sensors): \$2,500 - 2× High-performance servers: \$8,000 - Networking equipment (cables, routers): \$1,500 - **Subtotal:** \$18,000

1.10.4 9.4 Conference Travel

Target Conferences (4 years): - IEEE S&P, USENIX Security, CCS, NDSS (security top-tier) - IEEE INFOCOM, MobiCom (networking) - NeurIPS, ICML (ML venues for security) - Estimated: 4 major conferences × \$2,500/conference = \$10,000

1.10.5 9.5 Publications

- Open access fees: \$2,000-4,000 (2-3 papers)
- Page charges: \$500-1,000

1.10.6 9.6 Miscellaneous

- Books and resources: \$500
- Collaboration travel: \$1,500
- Contingency: \$2,000

1.10.7 9.7 Total Budget Estimate

Category	Amount
Computational Resources	\$15,000
Network Testbed Hardware	\$18,000
Software Licenses	\$2,500
Conference Travel	\$10,000
Publications	\$3,500
Miscellaneous	\$4,000
Total	\$53,000

Funding Sources: - University research assistantship - Department research grants
- External fellowships (NSF, industry) - Cloud provider research credits (AWS, Google)
- Industry partnerships (equipment donations)

1.11 10. Ethical Considerations

1.11.1 10.1 Data Privacy

- **Personal Data Protection:** Ensure all network traffic data is anonymized; remove personally identifiable information (PII)
- **GDPR Compliance:** Follow EU data protection regulations for data collection and storage
- **Consent:** Obtain informed consent for data collection in testbed experiments
- **Data Retention:** Implement data retention policies; delete data after research completion

1.11.2 10.2 Responsible Disclosure

- **Vulnerability Reporting:** Discovered vulnerabilities will be responsibly disclosed to affected vendors
- **Coordinated Disclosure:** Follow industry standards (90-day disclosure timeline)
- **No Exploit Development:** Research focuses on defense; no offensive exploit creation

1.11.3 10.3 Dual-Use Concerns

- **Defensive Purpose:** AI-IDS is designed purely for defensive cybersecurity

- **Misuse Prevention:** Document safeguards against malicious use
- **Access Control:** Restrict access to attack generation components (GANs for synthetic attacks)

1.11.4 10.4 Bias and Fairness

- **Dataset Bias:** Acknowledge and mitigate biases in training datasets
- **Fairness Evaluation:** Ensure IDS doesn't discriminate against specific users or traffic types
- **Diverse Testing:** Validate across diverse network environments and use cases

1.11.5 10.5 Environmental Impact

- **Energy Efficiency:** Optimize models to reduce computational energy consumption
- **Green Computing:** Use energy-efficient hardware and cloud providers with renewable energy
- **Carbon Footprint:** Track and report carbon footprint of large-scale experiments

1.11.6 10.6 Institutional Compliance

- **IRB Approval:** Obtain Institutional Review Board approval for human subjects research (user studies)
- **University Policies:** Comply with university research ethics policies
- **Export Control:** Ensure compliance with export control regulations for security research

1.12 11. Budget Estimate

1.12.1 Detailed Budget Breakdown

1.12.1.1 11.1 Personnel (if applicable)

- Research Assistant support: \$0 (covered by assistantship)
- Undergraduate research assistance: \$3,000 (summer work)

1.12.1.2 11.2 Equipment & Hardware (\$18,000)

Item	Quantity	Unit Cost	Total
OpenFlow SDN switches	5	\$1,000	\$5,000
High-performance servers	2	\$4,000	\$8,000
Raspberry Pi 4 (8GB)	10	\$100	\$1,000
IoT devices (mixed)	50	\$50	\$2,500
Network cables & accessories	-	-	\$500
Storage drives (backup)	2	\$500	\$1,000

1.12.1.3 11.3 Computational Resources (\$15,000)

Service	Description	Cost
AWS EC2 GPU instances	p3.8xlarge, 10,000 hours	\$12,000
Cloud storage	5TB S3/equivalent	\$1,500
Data transfer	Egress charges	\$1,500

1.12.1.4 11.4 Software & Licenses (\$2,500)

Software	Cost
Network analysis tools	\$1,500
Professional IDE licenses	\$500
Visualization software	\$500

1.12.1.5 11.5 Travel - Conferences (\$10,000)

Year	Conferences	Estimated Cost
Year 1	1 domestic conference	\$1,500
Year 2	1 international conference	\$3,000
Year 3	2 conferences (1 int'l, 1 domestic)	\$3,500
Year 4	1 conference (defense year)	\$2,000

1.12.1.6 11.6 Publications (\$3,500)

- Open access fees (2-3 papers): \$3,000
- Page charges: \$500

1.12.1.7 11.7 Miscellaneous (\$4,000)

- Books and resources: \$500
- Collaboration travel: \$1,500
- Printing and materials: \$300
- Contingency fund: \$1,700

1.12.1.8 11.8 Summary

Category	Amount	Percentage
Equipment & Hardware	\$18,000	34%
Computational Resources	\$15,000	28%
Conference Travel	\$10,000	19%
Publications	\$3,500	7%
Software & Licenses	\$2,500	5%
Personnel (undergrad support)	\$3,000	6%

Category	Amount	Percentage
Miscellaneous	\$1,000	2%
Total	\$53,000	100%

Funding Strategy: 1. **University Funding:** Research assistantship covers tuition and stipend 2. **Department Grants:** Apply for internal research grants (\$10K-15K) 3. **External Fellowships:** NSF GRFP, industry fellowships (\$30K-40K) 4. **Industry Partnerships:** Equipment donations, cloud credits (\$10K-20K) 5. **Cloud Credits:** AWS/Google Cloud research programs (\$5K-10K)

1.13 12. Conclusion

This PhD research proposal presents a comprehensive plan to develop an advanced AI-based intrusion detection system for next-generation networks. The proposed system addresses critical limitations of existing IDS through hybrid deep learning architectures, adaptive learning mechanisms, distributed deployment, and explainable AI integration.

1.13.1 Key Strengths of the Proposal

1. **Timely and Relevant:** Addresses urgent security challenges in 5G/6G, IoT, and edge computing
2. **Novel Approach:** Combines cutting-edge AI techniques (Transformers, RL, Federated Learning) in innovative ways
3. **Comprehensive Methodology:** Rigorous experimental design with multiple datasets and evaluation metrics
4. **Practical Impact:** Focus on deployable solutions with real-world validation
5. **Balanced Research:** Theoretical contributions complemented by practical implementation
6. **Feasible Plan:** Realistic 4-year timeline with clear milestones and deliverables

1.13.2 Expected Outcomes

By the completion of this PhD program, the research will deliver:

- **Academic Contributions:** 5-10 peer-reviewed publications in top conferences/journals
- **Practical Framework:** Open-source AI-IDS platform for industry adoption
- **Validated Solution:** Proven performance on benchmark datasets and real testbed
- **Community Impact:** Benchmark datasets, tools, and guidelines for future research
- **Trained Expert:** PhD graduate with deep expertise in AI-driven network security

1.13.3 Broader Impact

This research will advance the state-of-the-art in cybersecurity for next-generation networks, contributing to the protection of critical infrastructure, enterprise systems, and billions of IoT devices. The explainable and privacy-preserving nature of the proposed IDS addresses key concerns for industry adoption, while the open-source release will benefit the broader research community.

The interdisciplinary nature of this research—spanning artificial intelligence, cybersecurity, and networking—positions it to make significant contributions across multiple domains and prepare the candidate for leadership roles in academia or industry.

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End of PhD Research Proposal

This proposal represents a comprehensive research plan for developing advanced AI-based intrusion detection systems for next-generation networks. The proposed work addresses critical security challenges through innovative AI techniques, rigorous evaluation, and practical deployment considerations.

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